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MCA
(SEM IV) THEORY EXAMINATION 2022-23
COMPUTER GRAPHICS AND ANIMATION

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- a. Define Random Scan/Raster Scan displays.
- b. What is Aspect Ratio?
- c. What is Scaling?
- d. Distinguish between window and viewport.
- e. What is view plane normal vector?
- f. What is the use of control points?
- g. What are the types of reflection of incident light?
- h. Define shading.
- i. Define frame.
- j. Define complementary colors.

SECTION B

2. Attempt any three of the following: 10 x 3 = 30

- a. Calculate the pixel location approximating the first octant of a circle having centre at (4,5) and radius 4 units using Bresenham's algorithm.
- b. Explain the following composite transformations :
(i) Translation (ii) Rotation
- c. With suitable examples explain all 3D transformations.
- d. What is illumination? What are the various types of illumination.
- e. What is tweening? Explain in detail about motion tween with an example.

SECTION C

3. Attempt any one part of the following: 10 x 1 = 10

- a. Explain in detail about the Line drawing DDA scan conversion algorithm?
- b. Explain refresh CRT and color CRT monitors.

4. Attempt any one part of the following: 10 x 1 = 10

- a. Explain in detail about two dimensional viewing.
- b. Write about Cohen-Sutherland line clipping algorithm with an example.

5. Attempt any *one* part of the following:

10 x 1 = 10

- a. Write short notes on Bezier curve and spline.
- b. Discuss any one of the visible surface detection method.

6. Attempt any *one* part of the following:

10 x 1 = 10

- a. Explain in detail the CMY color model.
- b. Discuss in detail Gouraud shading.

7. Attempt any *one* part of the following:

10 x 1 = 10

- a. What is Animation? Describe the various animation techniques.
- b. What is storyboarding? Discuss in detail.

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